Numerical Methods: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

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Solution 1 — Bisection width. Return →

The width after n steps is $2/2^n$. We need $2/2^n < 10^{-3}$, so $2^n > 2000$. Since $2^{10} = 1024$ and $2^{11} = 2048$, **11 steps** are enough. **Check:** bisection gains one binary digit of accuracy per step.

Solution 2 — One Newton step. Return →

For $f(x) = x^2 - 2$, $f'(x) = 2x$. Thus $x_1 = 1.5 - (2.25 - 2)/3 = 1.41667$. The exact root is $\sqrt{2} = 1.41421$, so the error is about 0.00245. **Check:** one step reduces the error substantially because Newton's method is quadratic near the root, but this iterate is not yet accurate to three decimal places.

Solution 3 — Kepler's equation. Return →

At $E_0 = 1.0$, $f = 1.0 - 0.5\sin 1 - 1.0 = -0.4207$ and $f' = 1 - 0.5\cos 1 = 0.7297$. The Newton step gives $E_1 = 1.0 - (-0.4207)/0.7297 = 1.5766$ rad. The residual is $E_1 - 0.5\sin E_1 - 1.0 \approx 0.0766$. **Check:** the residual is much smaller than the initial residual, but more iterations are needed for high precision.

Solution 4 — Central difference. Return →

The estimate is $[(3.01)^2 - (2.99)^2]/0.02 = (9.0601 - 8.9401)/0.02 = 6.000$. The exact derivative is $2x = 6$. **Check:** the central difference is exact for a quadratic because the symmetric truncation error cancels.

Solution 5 — Trapezoidal rule. Return →

With $h = 0.5$, the estimate is $0.5[0.5f(0) + f(0.5) + 0.5f(1)] = 0.5[0 + 0.25 + 0.5] = 0.375$. The exact integral is $1/3 = 0.3333$. The trapezoidal estimate is high because x^2 is convex. **Check:** straight chords lie above a convex curve.

Solution 6 — Simpson's rule. Return →

Simpson's rule gives $(1/6)[f(0) + 4f(0.5) + f(1)] = (1/6)[0 + 1 + 1] = 1/3$. It is exact because Simpson's rule integrates quadratic polynomials exactly. **Check:** the method matches the analytic integral.

Solution 7 — Euler decay. Return →

Euler gives $y_1 = 1 - 0.1 = 0.9$ and $y_2 = 0.9 - 0.09 = 0.81$. The exact value is $e^{-0.2} = 0.8187$. Euler is low by about 0.0087. **Check:** Euler underestimates decay here because it uses the slope at the beginning of each step.

Solution 8 — Fixed-point iteration. Return →

Start with $E_0 = 0.5$. Then $E_1 = 0.5 + 0.2 \sin 0.5 = 0.5959$. Next $E_2 = 0.5 + 0.2 \sin 0.5959 = 0.6123$. **Check:** the iteration converges because $|e \cos E| < 1$ for $e = 0.2$.

Solution 9 — Convergence order. Return →

The errors are roughly squared at each step: $10^{-1} \to 10^{-2} \to 10^{-4} \to 10^{-8}$. This is quadratic convergence. Newton's method behaves this way near a simple root because the tangent approximation captures the function to first order, leaving an error proportional to the square of the previous error.

Solution 10 — Choose a method. Return →

A hybrid is best. Use a coarse search or bracket to locate the minimum or root robustly, then use Newton or a quasi-Newton method if derivatives are available and the model is smooth. If derivatives are

expensive, a secant or Brent-type method may be preferable. A safeguard is to reject steps that leave the bracket or increase the residual, and to verify convergence by changing the tolerance or initial guess.